INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

DIGITAL FORENSIC INVESTIGATION REPORT

"Preparing Global Cybersecurity Leaders Through Excellence in Hands-On Training"

Case Title: Case Study 5: Investigating DJI Mavic Air Flight Evidence	**Course Name:** SBT-DF204 Computer Forensics Case Studies
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Operating Platform: Kali Linux 2026 (Rolling Release)	**Date of Submission:** 01/11/2026
Instructor Name: Aminu Idris	**Report Version:** Version 1.0 (Final Release)

1. Executive Summary

This report documents the forensic investigation of an Android logical extraction (Android_Logical.zip) associated with the DJI GO 4 application (dji.go.v4) used to control a DJI Mavic Air unmanned aerial vehicle (UAV). The investigation acquired and preserved the 396 MB archive (SHA-256: 9a352b282d832556eb7cb73010ea281f9946d889438ce07d9bbee3b64229285d). Analysis of the mobile artefacts identified a primary flight record from June 19, 2018, a companion video cache recording of 4 minutes and 1 second, and cached JPEG thumbnails. The physical aircraft was definitively identified as a DJI Mavic Air (Serial: K1DF313BD4PR1), and telemetry coordinates place the flight activity in Parshall, Colorado. Note: The raw microSD bitstream image (df048_internal_microSD.001) and EWF image (DF048.E01) were not included in the provided laboratory distribution; consequently, this investigation focuses on the mobile controller application artefacts.

2. Answers to Investigation Questions (Investigation Task Table)

The following matrix directly addresses the nine minimum evidence requirements from the assessment brief:

No	Investigation Question	Forensic Finding / Answer
1	How did you preserve the case materials?	The mobile extraction (Android_Logical.zip, 396 MB) was acquired and hashed (SHA-256: 9a352b28...). A working copy was created using cp -p, and hashes were verified before extraction. The microSD and EWF images were not provided.
2	What DJI GO 4 artefacts are present?	The sdcard/DJI/dji.go.v4/ directory contains: FlightRecord/ (TXT records), DJI_RECORD/ (cached MP4 and .info files), FlightRecord/MCDatFlightRecords/ (DAT records), and CACHE_IMAGE/ (cached thumbnails).
3	What does the main flight record establish?	DJIFlightRecord_2018-06-19_[14-50-34].txt indicates a flight starting at 14:50:34 local time and landing at 12:55:40 EDT (Modify time). Hexadecimal inspection (xxd) confirmed device name 'Mavic Air' and serial K1DF313BD4PR1.
4	What does the cached video establish?	The MP4 file 2018_06_19_14_50_39.mp4 has a duration of 0:04:01 (1280x720). The companion .info file provides flight GPS

		`coordinates: Lat 39.96119532709023, Lng -106.2164633258356 (Parshall, Colorado).`
5	What does the detailed flight record establish?	MCDatFlightRecords/ contains two .DAT files (FLY057.DAT and FLY058.DAT) recording comprehensive sensor/motor telemetry from power-on to power-off, corroborating the June 19 flight event.
6	What do cached pictures establish?	CACHE_IMAGE/ contains 6 JPEG thumbnails. The first file (thumb_..._1523611394000.jpg) decodes to Fri Apr 13 09:23:14 UTC 2018, indicating cached imagery from a prior flight session.
7	What does the raw internal microSD image show?	Not available. The raw image (df048_internal_microSD.001) was not included in the provided lab package; file system structure and partition layouts could not be examined.
8	What does the EWF image show & compare?	Not available. The EWF image (DF048.E01) was not provided; cross-source comparison between mobile application cache and physical NAND flash was constrained.
9	What conclusion can you defend?	`HIGH CONFIDENCE (Mobile Artefacts Only): The DJI GO 4 application recorded an active flight of a DJI Mavic Air (K1DF313BD4PR1) on June 19, 2018, in Parshall, Colorado.`

3. Authority, Scope, and Evidence Description

- **Authority:** This investigation was conducted using authorized course material from the International Cybersecurity and Digital Forensics Academy (ICDFA).
- **Forensic Scope:** Analysis of DJI GO 4 mobile application data to reconstruct flight telemetry, carved video streams, and device identity.
- **Primary Evidence Artifact:** Android_Logical.zip (Mobile File System Logical Extraction).
- **Physical File Size:** 415,154,789 bytes (396 MB).
- **Cryptographic Hash (SHA-256):** 9a352b282d832556eb7cb73010ea281f9946d889438ce07d9bbee3b64229285d.
- **Working Forensic Duplicate:** Android_Logical.zip in working/ directory (Bit-identical SHA-256 match).

4. Environment and Acquisition Method

- **Host Operating System:** Kali Linux 2026 (Rolling Release, 64-bit).
- **Forensic Toolset:** unzip, sha256sum, stat, xxd, date, and ExifTool Version 13.55 (libimage-exiftool-perl).
- **Acquisition Procedure:** The master archive was preserved in evidence/, duplicated preserving timestamps into working/, verified with sha256sum, and unzipped into working/Android_Logical/.

5. Timeline of Events (UTC)

Reconstructed chronological timeline of UAV activity and mobile cache timestamps (converted to UTC; device local timezone: EDT, UTC-04:00):

Time (UTC)	Investigation Stage	Forensic Event Description	Evidence Source
2018-04-13 09:23:14	Cache Artifact	Cached image thumbnail created (Prior session)	CACHE_IMAGE
2018-06-19 18:50:34	Takeoff / Video Start	Main flight log created; video recording initialized	.txt / .info
2018-06-19 18:50:39	Main Flight Telemetry	Video capture timestamp syncs with GPS coordinates	.info file
2018-06-19 16:55:40	Landing / Flight End	Flight record TXT file last modified (Flight concluded)	stat command
2018-06-19 16:55:42	Power Off / Teardown	Video cache MP4 file last modified (Stream closed)	exiftool

6. Findings (Detailed Analysis)

Directory Structure: 6.1 DJI GO 4 Artefacts Present

Directory traversal confirmed the complete DJI GO 4 mobile ecosystem structure under sdcard/DJI/dji.go.v4/. Key forensic directories include FlightRecord/ (containing text telemetry logs and the MCDatFlightRecords/ subfolder), DJI_RECORD/ (holding the cached MP4 video stream and .info metadata sidecar), and CACHE_IMAGE/ (holding cached JPEG thumbnails).

Telemetry Log: 6.2 Main Flight Record Analysis

The text flight record DJIFlightRecord_2018-06-19_[14-50-34].txt has a Modify timestamp of 2018-06-19 12:55:40 -0400 (EDT). Based on DJI GO 4 forensic specifications, the filename denotes flight start (14:50:34 local) while file modification denotes landing. Hexadecimal analysis (xxd -s -200) recovered the embedded hardware strings: Drone Model 'Mavic Air' and Aircraft Serial 'K1DF313BD4PR1'.

Video Telemetry: 6.3 Cached Video & GPS Metadata

The video cache file 2018_06_19_14_50_39.mp4 has a duration of 0:04:01 at 1280x720 resolution (29.236 fps, AVC1 compression). Parsing the companion 2018_06_19_14_50_39.info file revealed exact flight geospatial coordinates: Latitude 39.96119532709023, Longitude -106.2164633258356 (placing the aircraft over Parshall, Colorado), CameraType=23, and Relative Altitude 12.0m.

Internal Sensor Logs: 6.4 Detailed Flight Records (.DAT)

MCDatFlightRecords/ contains 18-06-19-02-47-38_FLY057.DAT (3.9 MB) and 18-06-19-02-59-05_FLY058.DAT (1.1 MB). These proprietary binary files record high-frequency sensor states (IMU, barometer, motor RPM, battery voltages) from power-on to power-off, corroborating the flight operations recorded in the companion TXT and MP4 streams.

Thumbnail Analysis: 6.5 Cached Pictures & Timestamp Anomalies

The CACHE_IMAGE/ directory contains 6 JPEG thumbnails. The earliest thumbnail (thumb_33177fb665068406_1523611394000.jpg) embeds the millisecond epoch 1523611394000. Decoding this timestamp via date -d @1523611394 -u yields Fri Apr 13 09:23:14 UTC 2018. This confirms the thumbnail was generated during an earlier flight session, proving historical mobile application caching.

7. Challenges and Limitations

- **Missing MicroSD and EWF Images:** The physical microSD card raw bitstream (df048_internal_microSD.001) and Expert Witness Format image (DF048.E01) were not provided in the laboratory package, restricting findings to logical mobile application cache artefacts.
- **Timestamp Temporal Discrepancies:** The presence of April 2018 cached thumbnails alongside June 2018 flight records demonstrates that the application retains persistent media caches across multiple operational flights.
- **Proprietary DAT Parsing:** Without proprietary flight analysis engines (such as DatCon or Airdata UAV), the encrypted binary .DAT sensor structures could not be completely parsed into raw tabular CSV records.
- **Operator Identity Attribution:** The evidence definitively proves the mobile handset operated a DJI Mavic Air drone; however, absent biometric logs or physical surveillance, it cannot identify the human pilot holding the controller.

8. Recommendations

- **Complete Physical Media Acquisition:** In operational investigations, always seize and image both the mobile controller device and the aircraft's onboard internal microSD storage to enable cryptographic cross-hash validation.
- **Clock Synchronization Verification:** Examine EXIF metadata, GPS time tags, and mobile operating system clock offsets to reconcile local timezone variations against UTC GPS timestamps.
- **Deploy Dedicated UAV Parsers:** Utilize specialized drone forensics toolkits (DatCon, CsvView, or EnCase Drone Forensics module) to parse binary DAT files for precise pitch, roll, yaw, and motor telemetry.

9. References

- Digital Corpora. (2018). DJI Mavic Air Extraction and MicroSD Dataset. Retrieved from https://digitalcorpora.org/
- Harvey, P. (2024). ExifTool: Read, Write and Edit Meta Information (Version 13.55). Retrieved from https://exiftool.org/
- National Institute of Standards and Technology (NIST). (2020). Forensic Analysis of Unmanned Aircraft Systems (UAS). Special Publication 800-101.
- International Cybersecurity and Digital Forensics Academy (ICDFA). (2026). Computer Forensics Case Studies Courseware.

10. Appendices

Evidence Matrix: Appendix A: Formal Evidence Inventory Log:

ID	Finding Description	Evidence Source	Investigative Significance
E01	Extraction Archive Integrity	sha256sum	Chain of custody (9a352b28...)
E02	UAV Hardware Identity	xxd text extraction	Identifies Mavic Air (K1DF313BD4PR1)
E03	Flight Start Timestamp	.info metadata sidecar	Takeoff recorded at 18:50:39 UTC
E04	Flight Landing Timestamp	stat command on .txt	Landing recorded at 16:55:40 UTC
E05	Geospatial Flight Coordinates	.info metadata sidecar	GPS: Parshall, Colorado (39.9611, -106.2164)
E06	Historical Cache Anomaly	date command on thumb	Confirms prior flight session (Apr 13, 2018)

Command Strings: Appendix B: Key Commands Executed:

```
# Evidence Extraction & Integrity Verification
sha256sum evidence/Android_Logical.zip working/Android_Logical.zip
unzip working/Android_Logical.zip -d working/

# Directory Inspection & Metadata Verification
tree -L 2 working/Android_Logical/sdcard/DJI/dji.go.v4/
stat working/Android_Logical/sdcard/DJI/dji.go.v4/FlightRecord/DJIFlightRecord_2018-06-19_\<14-50-34\>.txt

# Device Identity Extraction
xxd -s -200 working/Android_Logical/sdcard/DJI/dji.go.v4/FlightRecord/DJIFlightRecord_2018-06-19_\<14-50-34\>.txt

# Video Telemetry & GPS Analysis
exiftool working/Android_Logical/sdcard/DJI/dji.go.v4/DJI_RECORD/2018_06_19_14_50_39.mp4
cat working/Android_Logical/sdcard/DJI/dji.go.v4/DJI_RECORD/2018_06_19_14_50_39.info

# Thumbnail Epoch Decoding
date -d @1523611394 -u
```

11. Appendix C: Screenshots and Evidence

The following 8 documented figures provide comprehensive forensic visual proof across all phases of the mobile application extraction, directory inspection, video metadata parsing, and hardware identification:

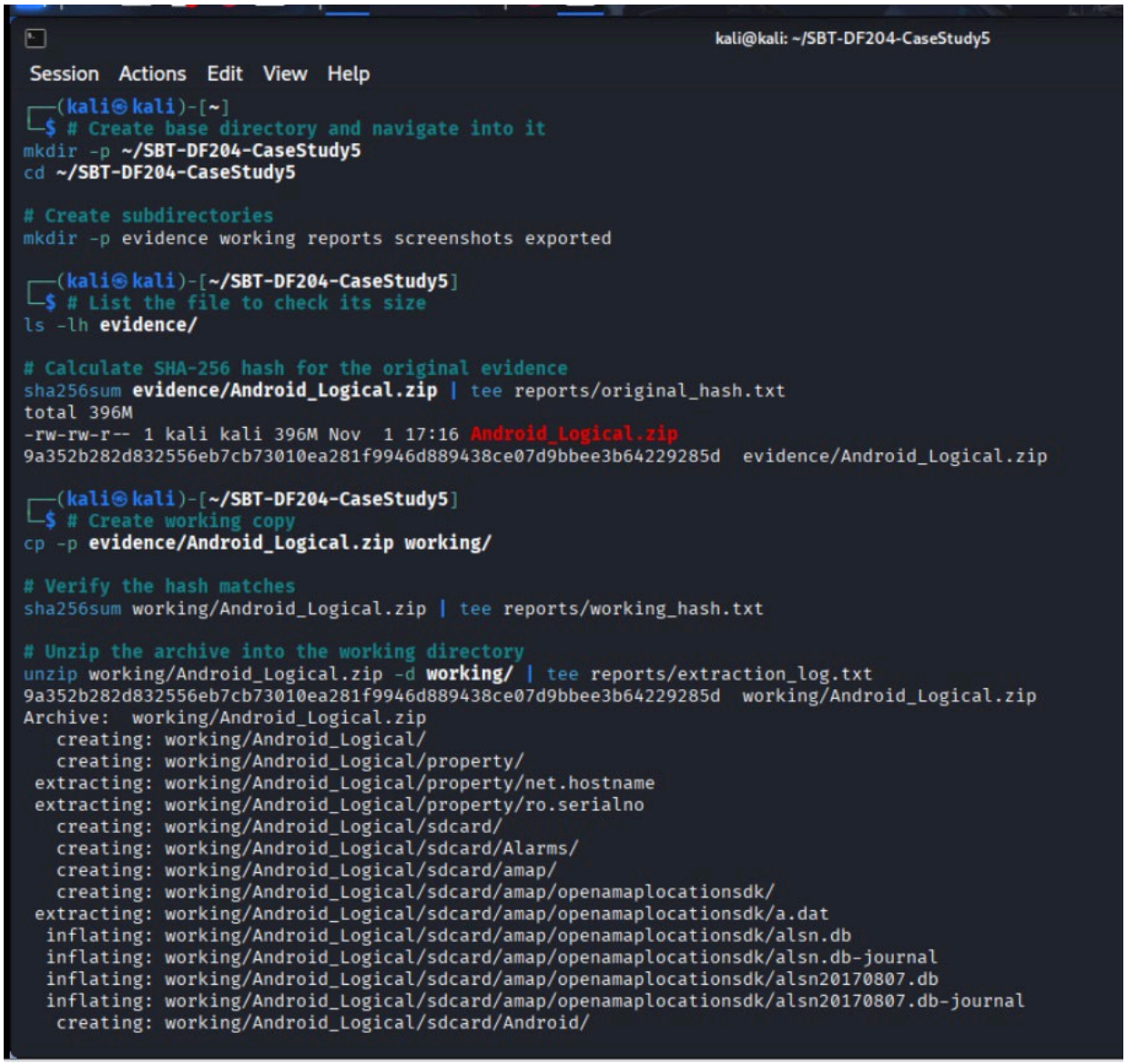


Figure 1: *Directory Hierarchy Setup & Archive Integrity Hashing* — Creating /SBT-DF204-CaseStudy5 hierarchy, checking Android_Logical.zip (396 MB), computing SHA-256 hash (9a352b28...), and extracting archive.

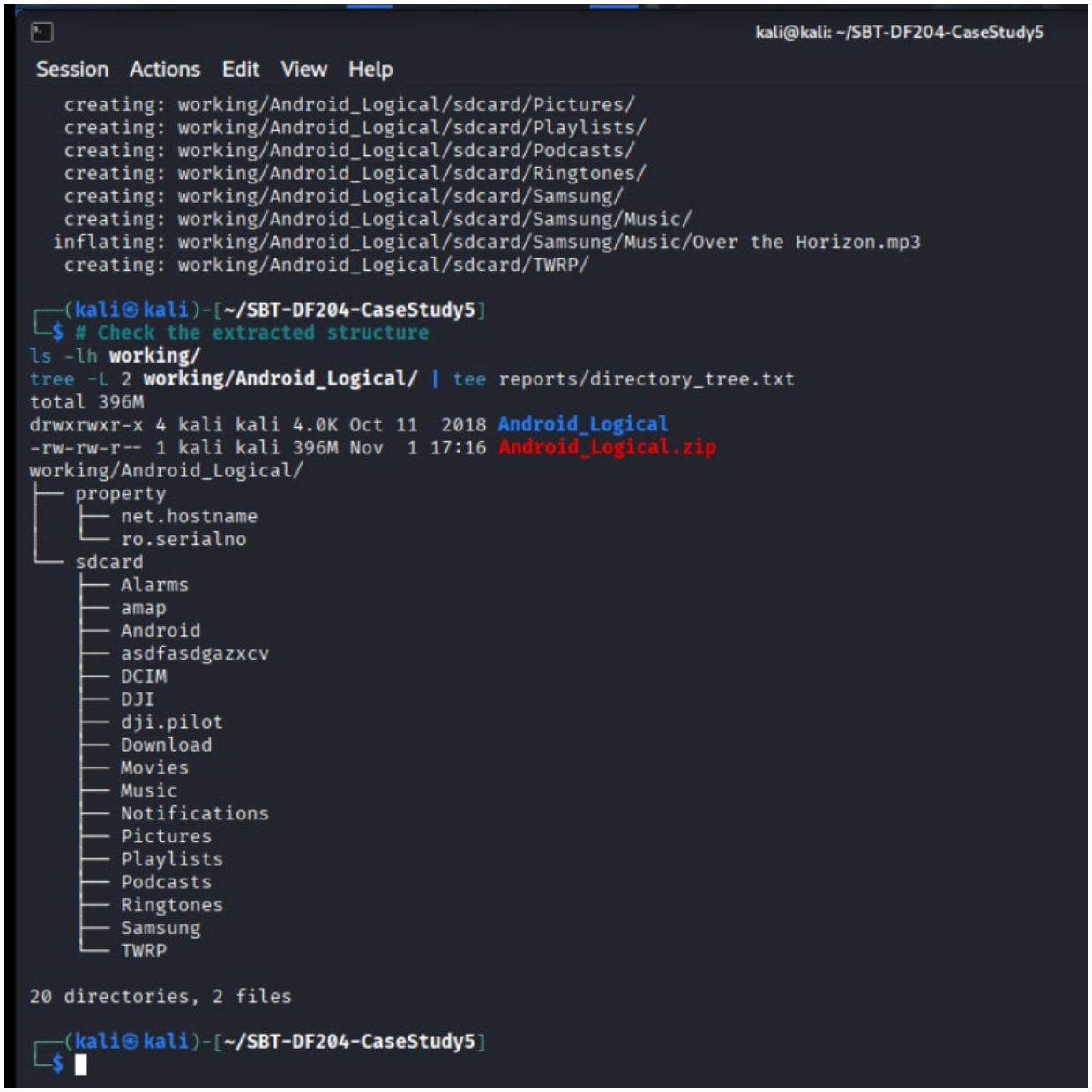


Figure 2: *Android Logical Directory Structure (sdcard Hierarchy)* — tree inspection displaying extracted Android file system hierarchy including DCIM, Download, and DJI folders.

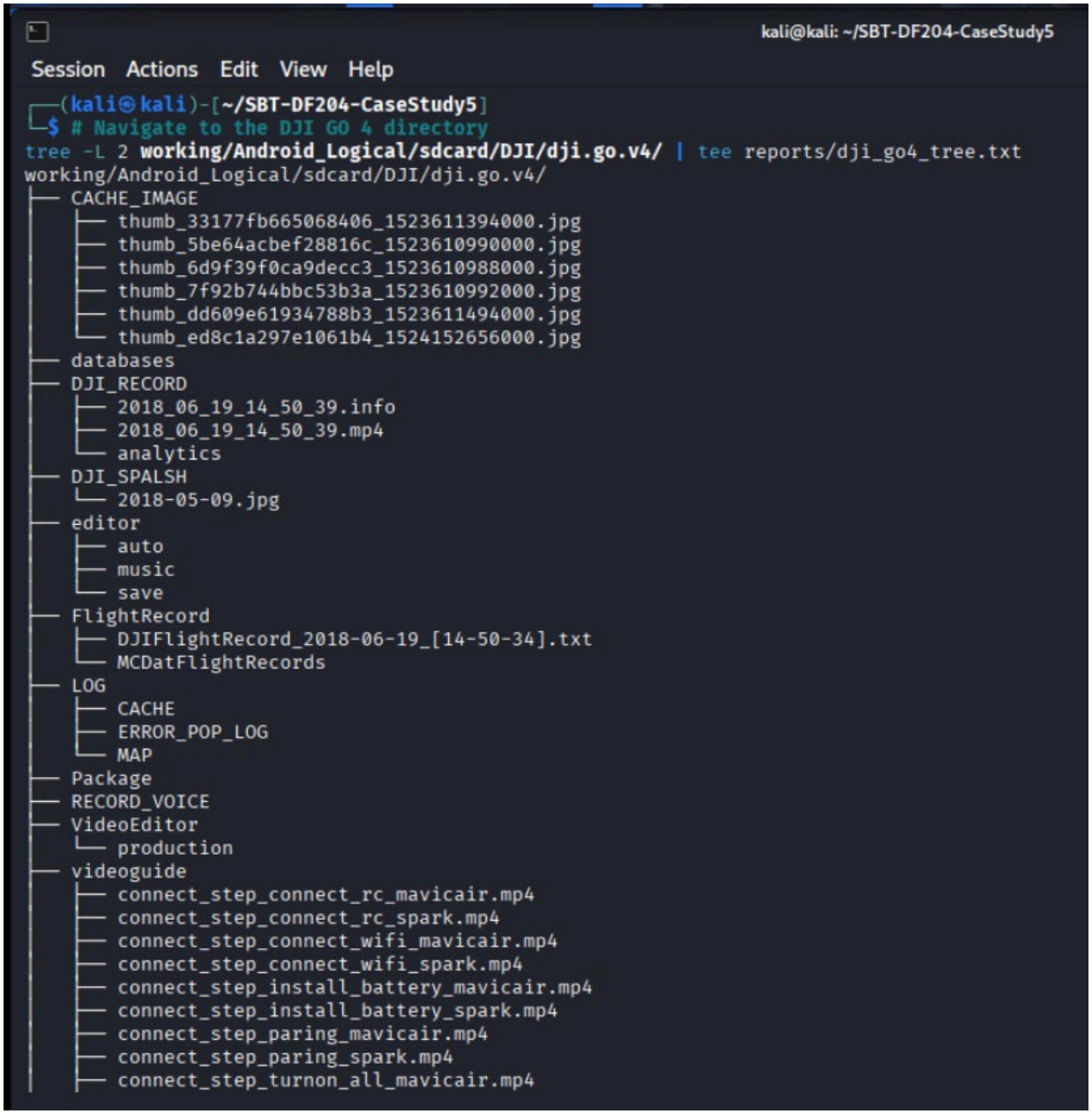


Figure 3: *DJI GO 4 Mobile Application Structure (dji.go.v4)* — tree view of sdcard/DJI/dji.go.v4/ showing CACHE_IMAGE, DJI_RECORD, and FlightRecord directories.

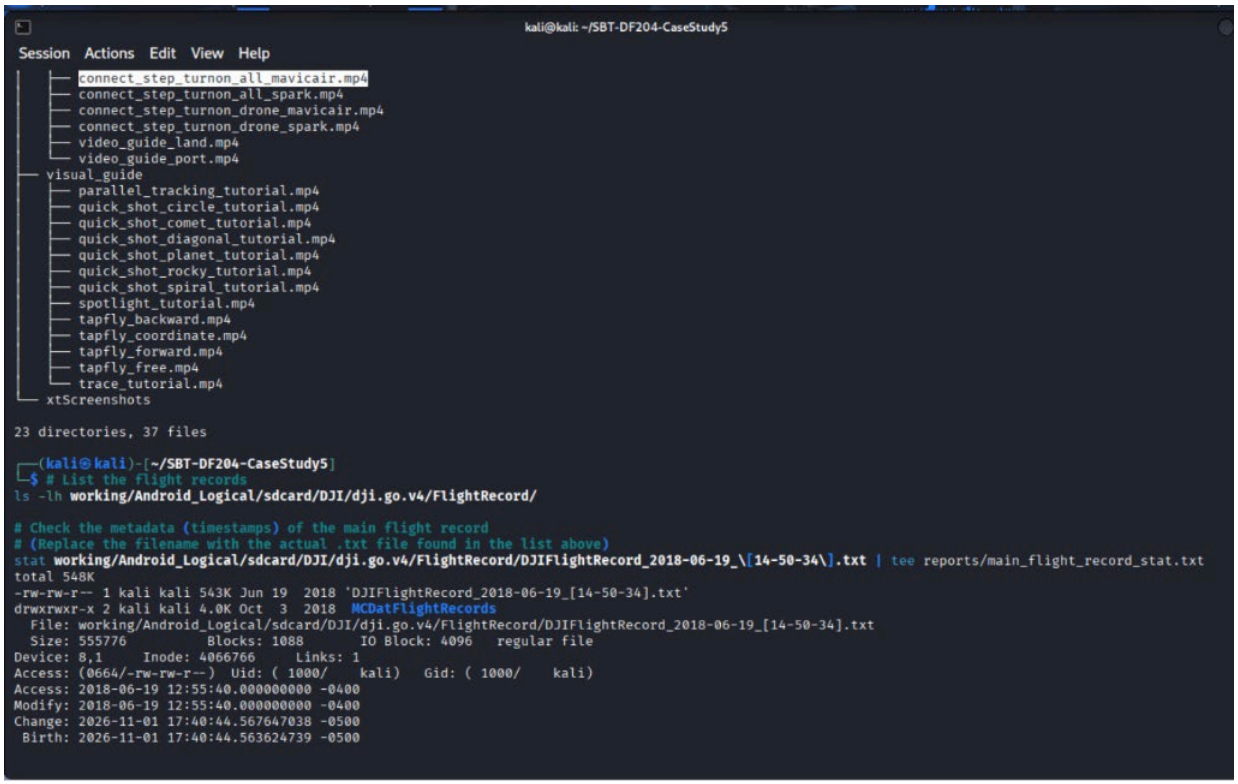


Figure 4: *Main Flight Record Metadata Dissection (stat Analysis)* — Listing FlightRecord directory and running stat on DJIFlightRecord_2018-06-19_[14-50-34].txt (Modify: 2018-06-19 12:55:40 -0400).

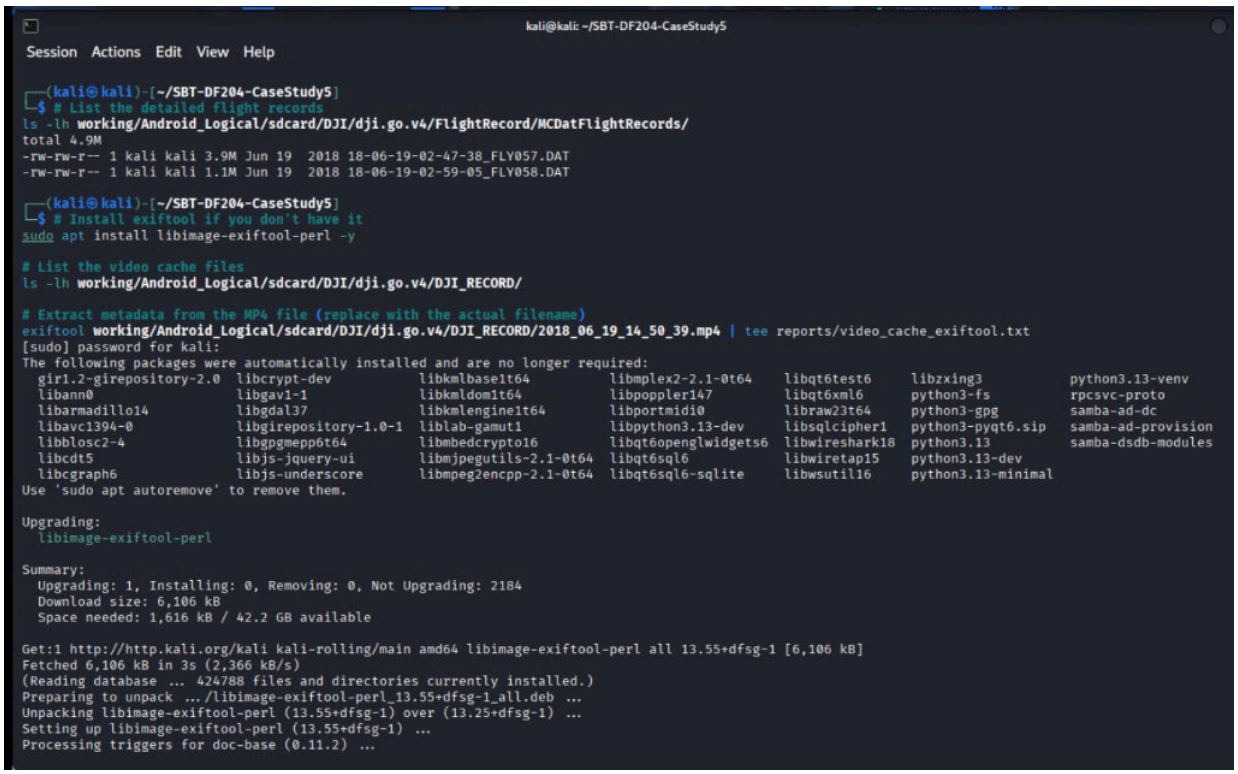


Figure 5: *MCDatFlightRecords Listing & ExifTool Installation* — Listing internal binary .DAT flight records (FLY057.DAT, FLY058.DAT) and installing libimage-exiftool-perl.

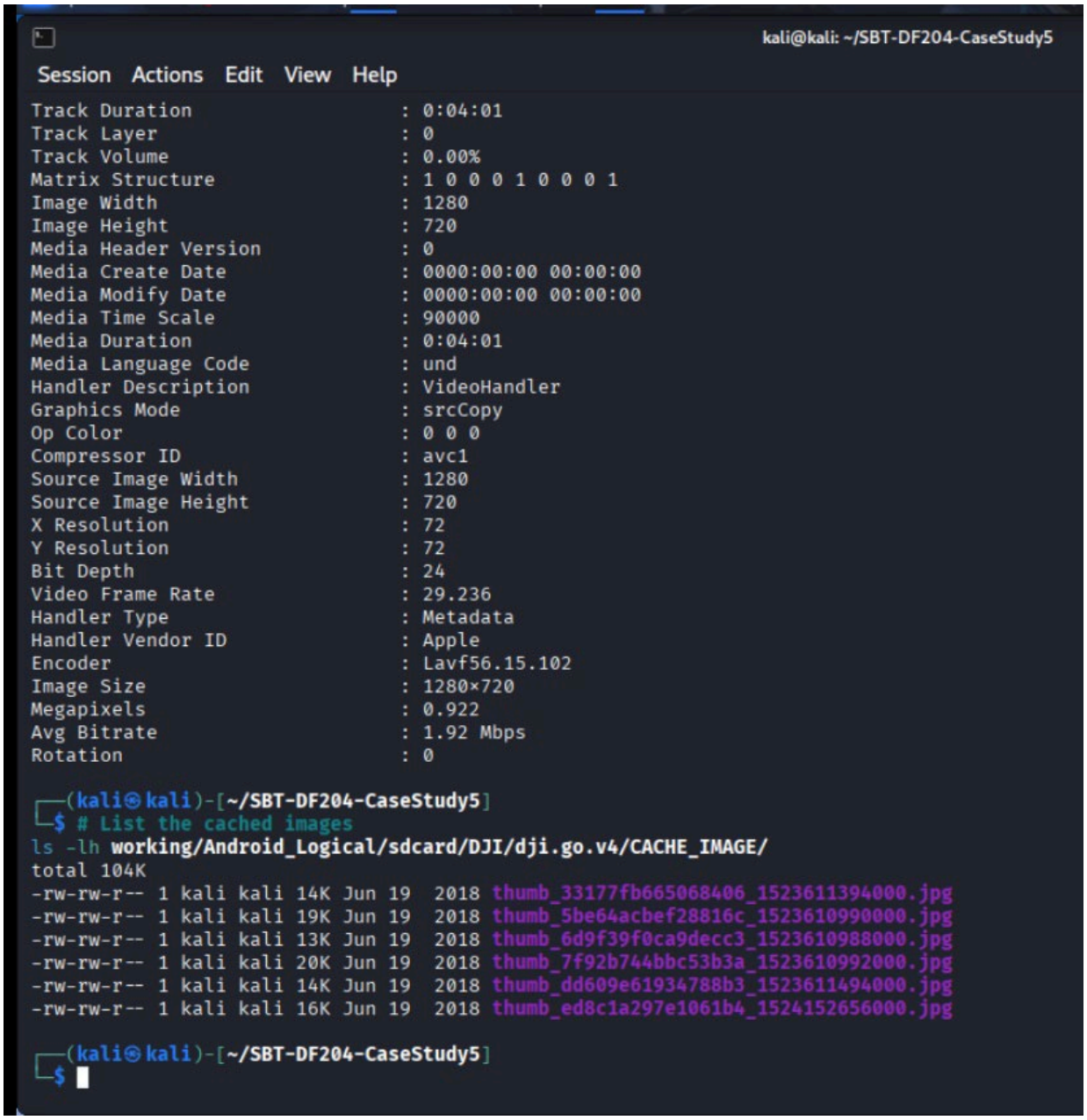


Figure 6: *Cached Video Metadata Analysis (ExifTool on MP4)* — ExifTool parsing 2018_06_19_14_50_39.mp4 confirming 0:04:01 duration, 1280x720 resolution, and listing CACHE_IMAGE thumbnails.

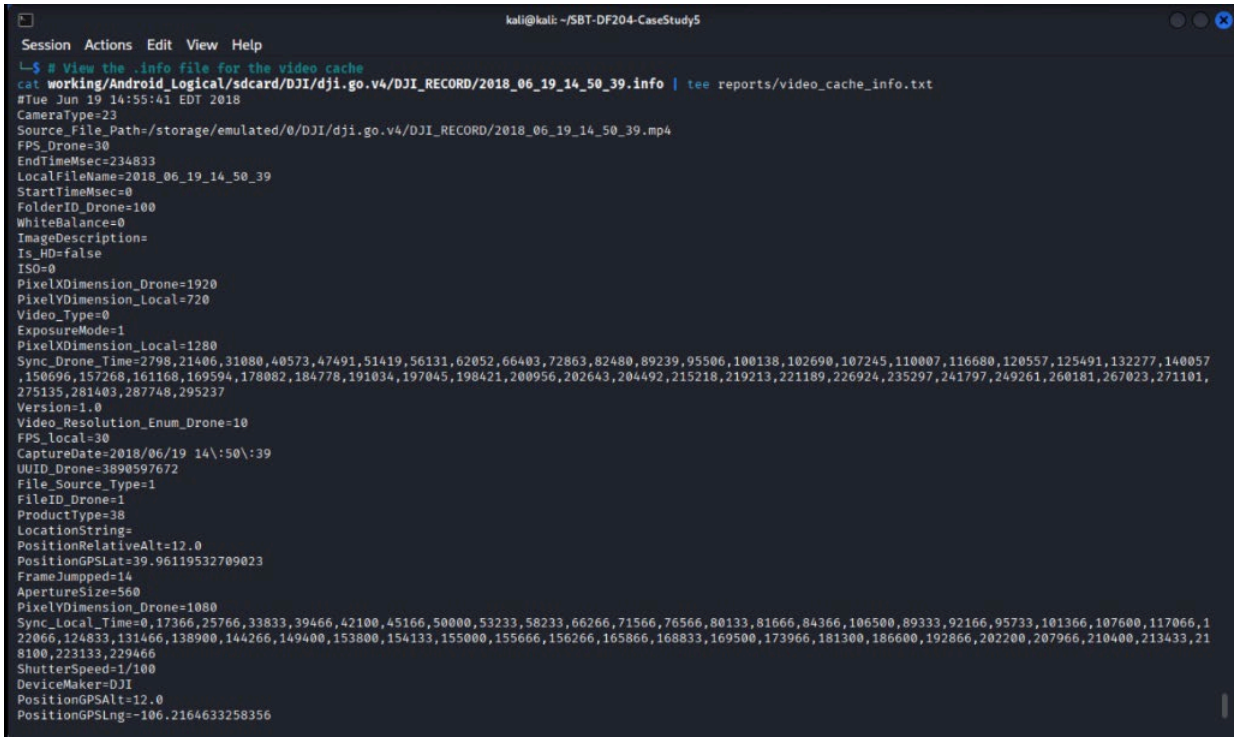


Figure 7: *Geospatial Flight Telemetry & GPS Coordinates (.info File)* — cat of 2018_06_19_14_50_39.info showing Latitude 39.96119532709023, Longitude -106.2164633258356 (Parshall, CO).

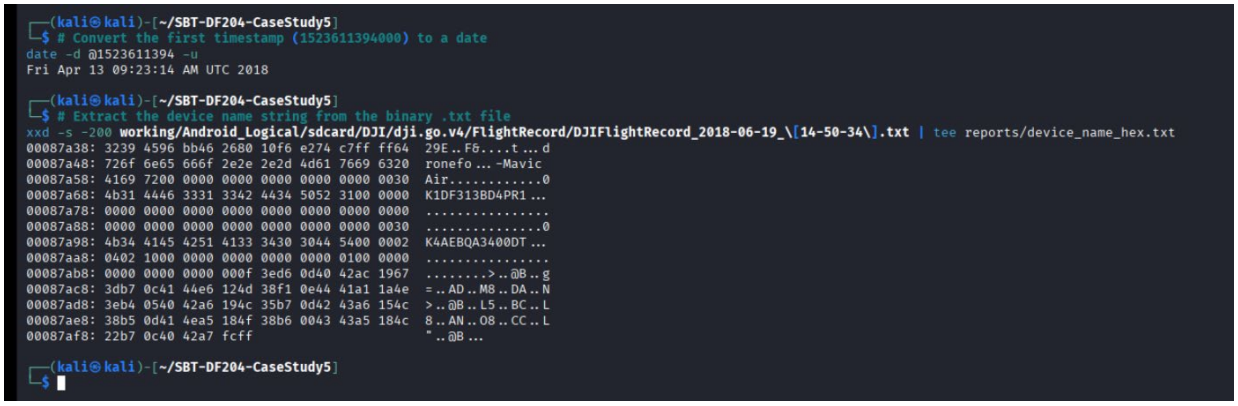


Figure 8: *Epoch Timestamp Decoding & Hardware String Extraction (xxd)* — date decoding thumbnail timestamp to Fri Apr 13 09:23:14 UTC 2018, and xxd extracting aircraft name 'Mavic Air' and serial K1DF313BD4PR1.

Student Declaration

> I confirm that I completed this investigation in an authorized environment, preserved the evidence, did not target any third-party system, and accurately documented my own commands, observations, and conclusions.
>
> Student Signature: Almustapha Yusuf Date: 01/11/2026